Dobrodošli v Tidyverse

Tomaž Skrbinšek

26. 10. 2020

Nekateri materiali v prezentaciji so prilagojeni po
https://github.com/rstudio-education/master-the-tidyverse/
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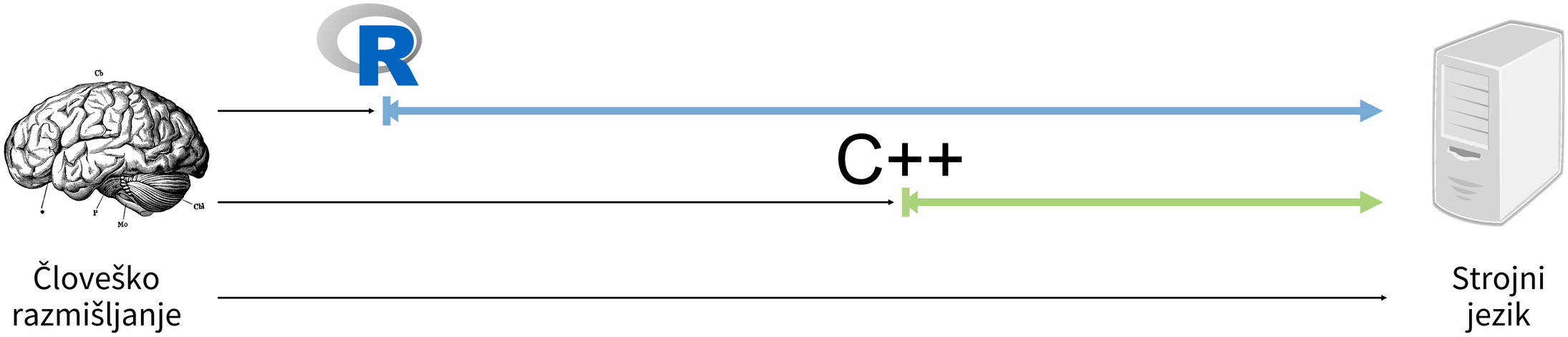
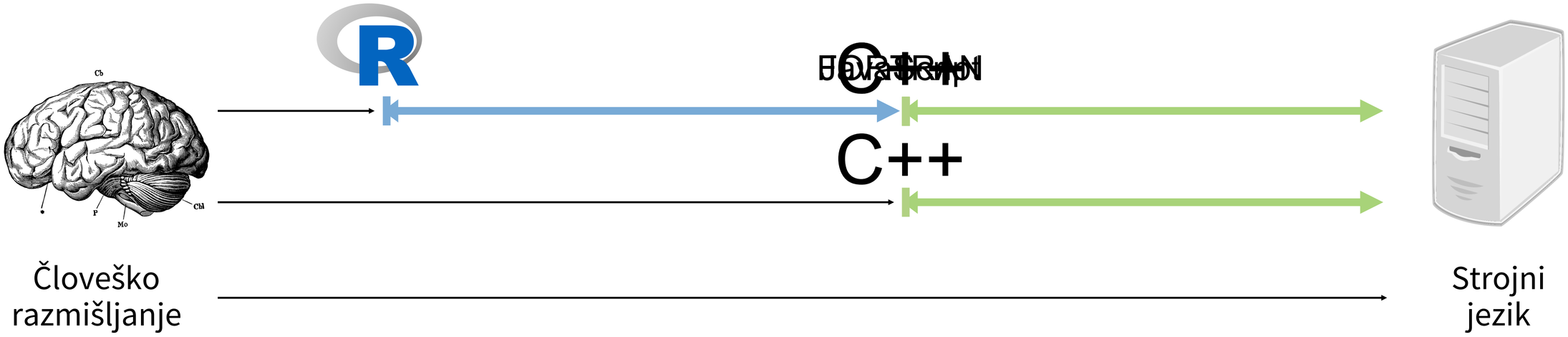
"Data Science" – podatkovna znanost

Zbiranje podatkov
Uvoz podatkov v software
Urejanje podatkov v uporabno obliko
Raziskovanje, testi
Transformacija podatkov
Pisanje kode algoritmov za modeliranje
Vizualizacija podatkov in/ali rezultatov
Komunikacija rezultatov
Pisanje članka ali aplikacije
Objava članka ali aplikacije
Oblikovanje hipotez
Načrtovanje eksperimenta
R

R – Računalniški jezik za znanstvenike



R – Računalniški jezik za znanstvenike



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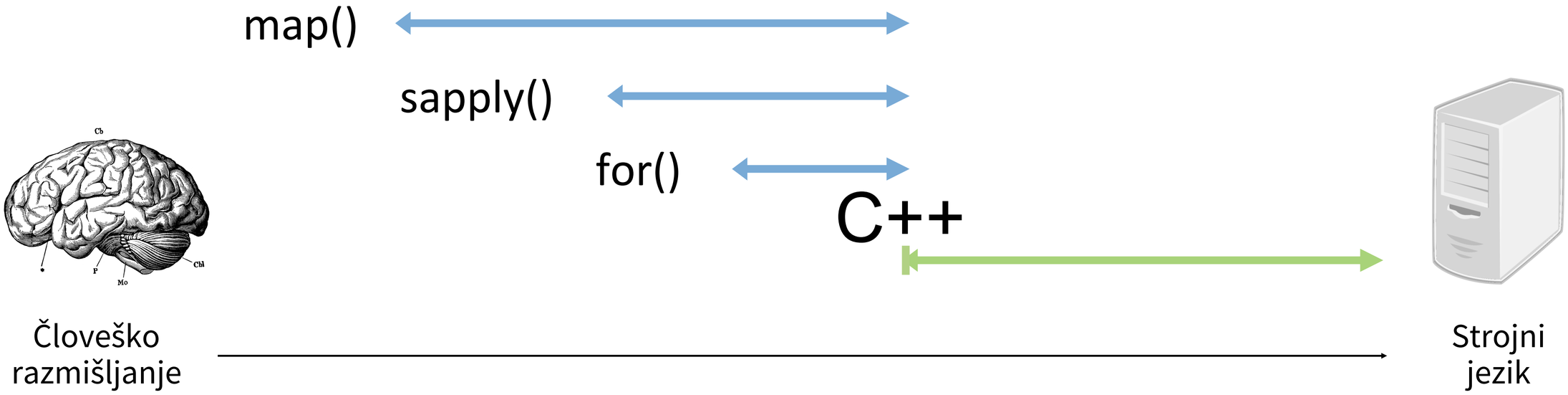
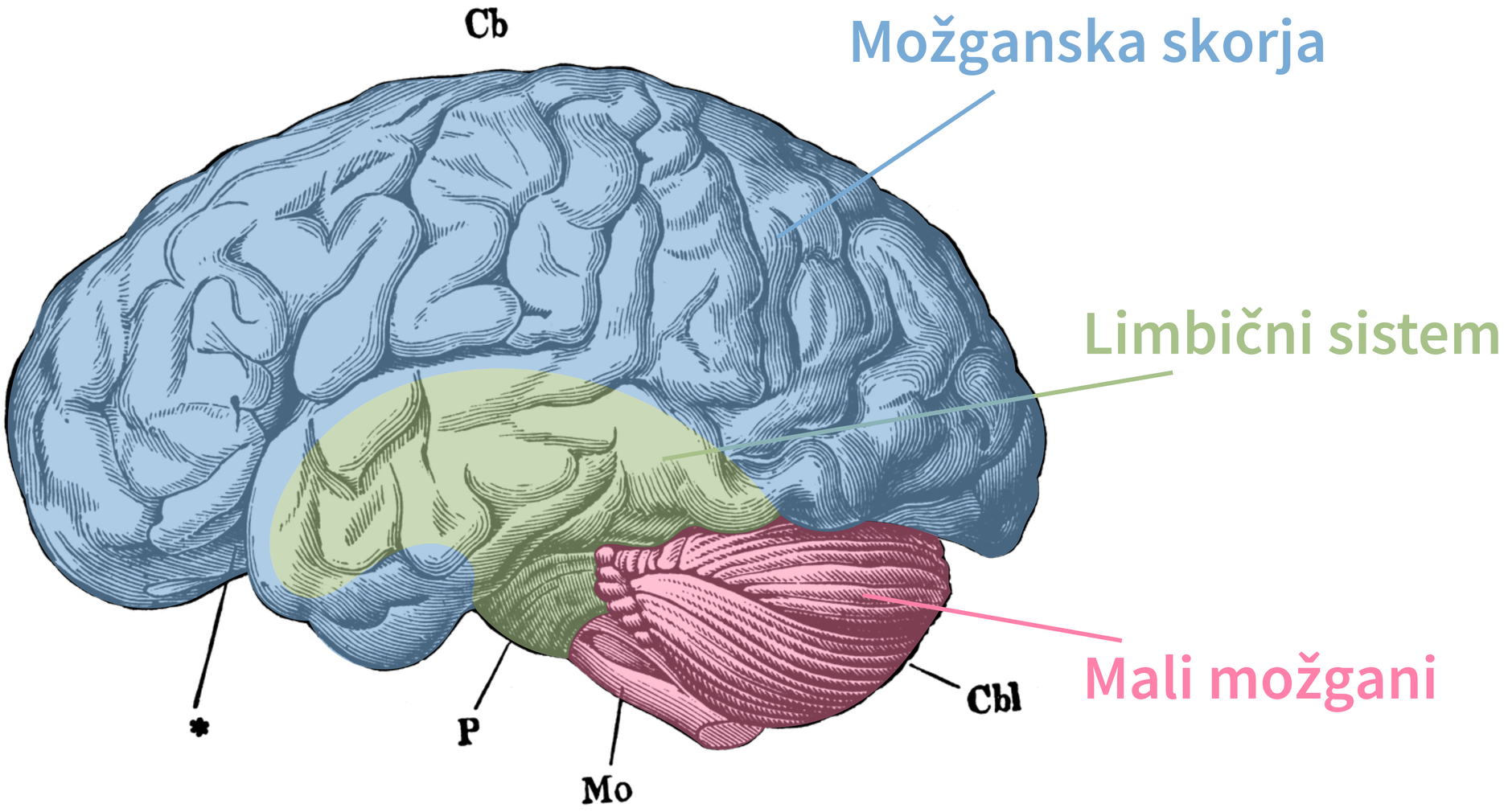
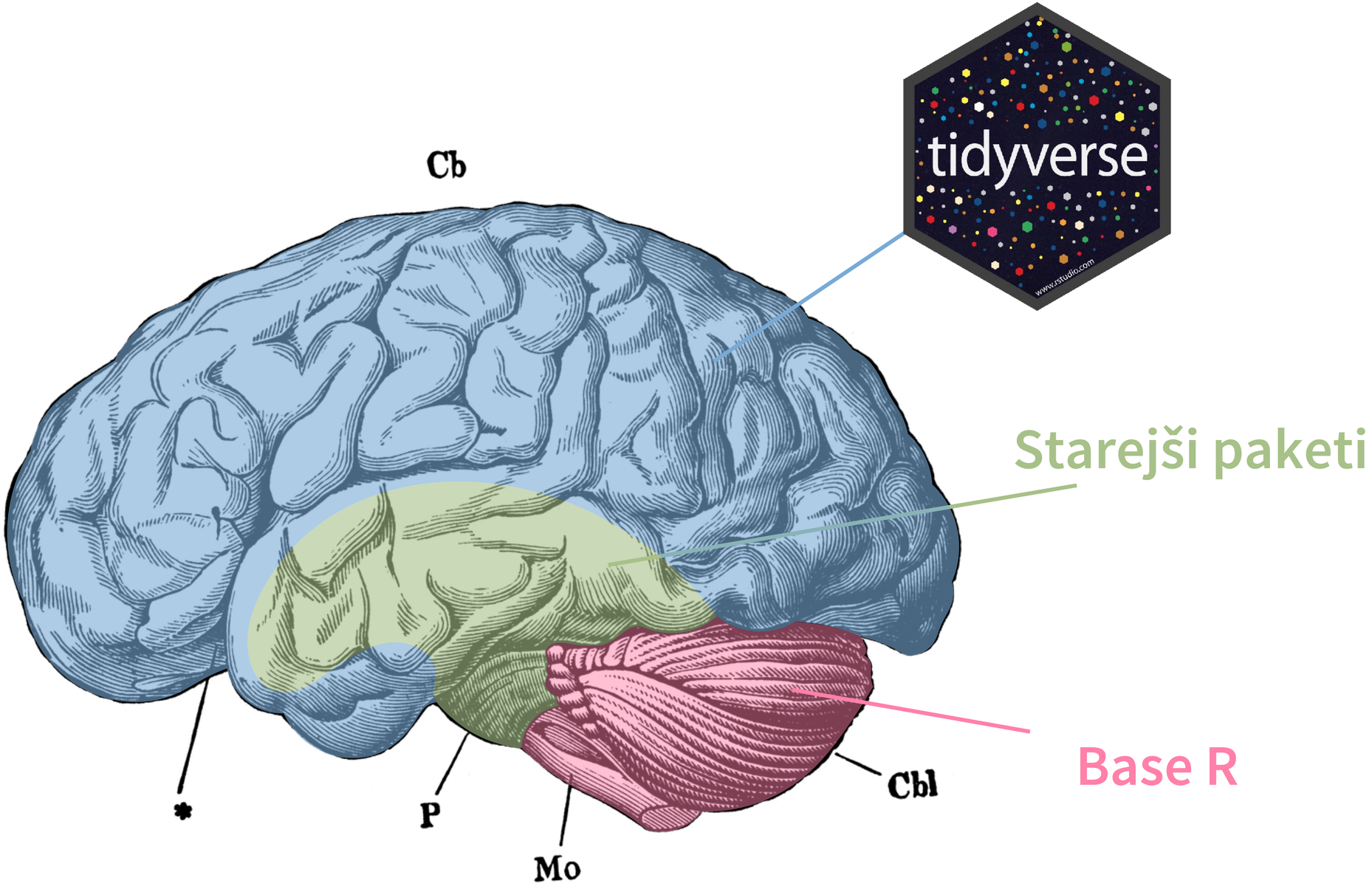
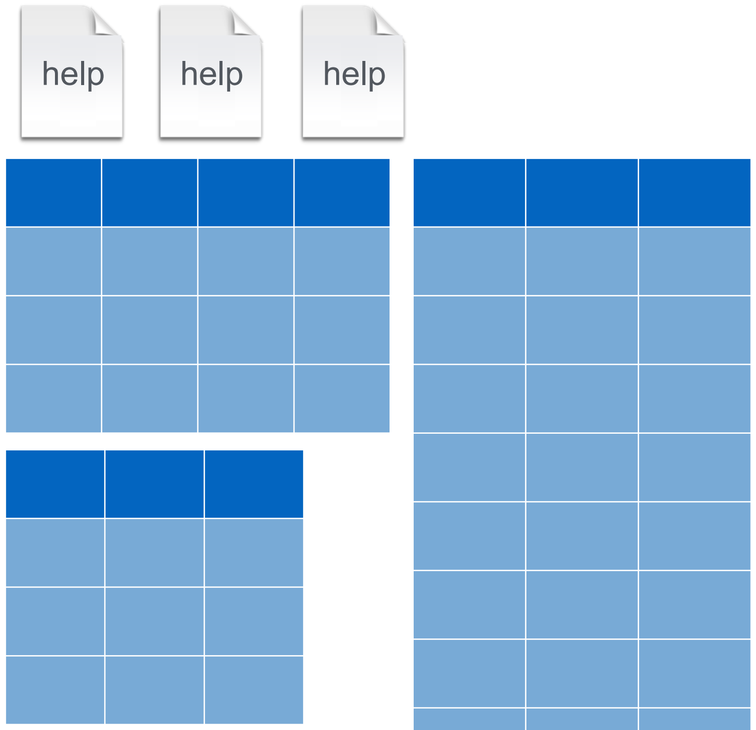



Cb
Možganska skorja
Limbični sistem
Mali možgani
Cbl
*
P
Mo

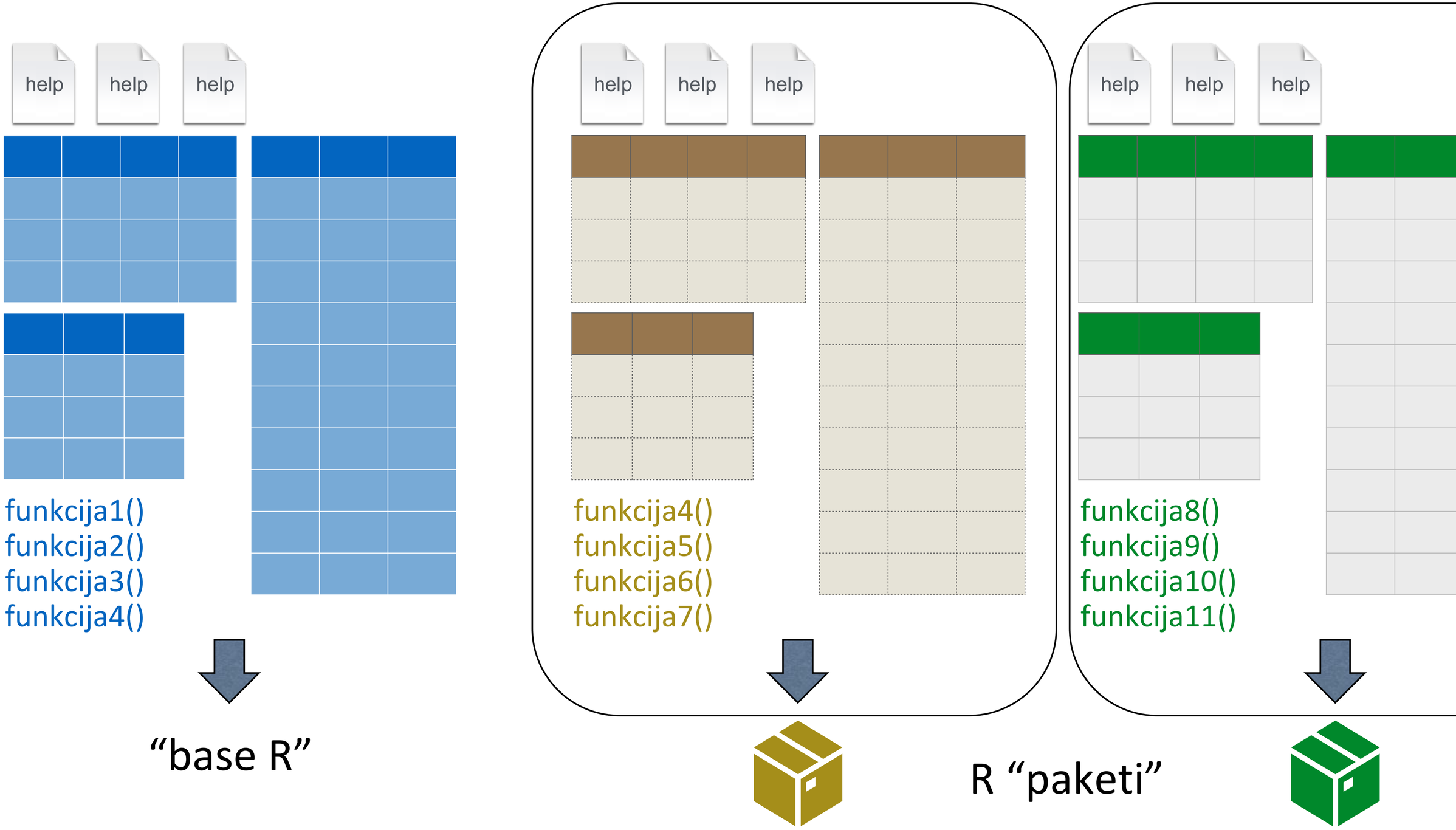

tidyverse
www.rstudio.com
Cb
Starejši paketi
Base R
Cbl
*
P
Mo

R Paketi



funkcija1()
funkcija2()
funkcija3()
funkcija4()

Base R: funkcije, podatki, pravila in sistem pomoči


help
help
help
funkcija1()
funkcija2()
funkcija3()
funkcija4()
"base R"
help
help
help
funkcija4()
funkcija5()
funkcija6()
funkcija7()
help
help
help
funkcija8()
funkcija9()
funkcija10()
funkcija11()
R "paketi"

The Comprehensive R Archive

Secure https://cran.r-project.org

CRAN
Mirrors
What's new?
Task Views
Search

About R
R Homepage
The R Journal

Software
R Sources
R Binaries
Packages
Other

Documentation
Manuals
FAQs
Contributed

Available CRAN Packages By Name

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

A3	Accurate, Adaptable, and Accessible Error Metrics for Predictive Models
abbyyR	Access to Abbyy Optical Character Recognition (OCR) API
abc	Tools for Approximate Bayesian Computation (ABC)
ABCanalysis	Computed ABC Analysis
abc.data	Data Only: Tools for Approximate Bayesian Computation (ABC)
abcdeFBA	ABCDE_FBA: A-Biologist-Can-Do-Everything of Flux Balance Analysis with this package
ABCoptim	Implementation of Artificial Bee Colony (ABC) Optimization
ABCp2	Approximate Bayesian Computational Model for Estimating P2
ABC.RAP	Array Based CpG Region Analysis Pipeline
abcrf	Approximate Bayesian Computation via Random Forests
abctools	Tools for ABC Analyses
abd	The Analysis of Biological Data
abf2	Load Gap-Free Axon ABF2 Files
ABHgenotypeR	Easy Visualization of ABH Genotypes
abind	Combine Multidimensional Arrays
abjutils	Useful Tools for Jurimetrical Analysis Used by the Brazilian Jurimetrics Association
abn	Modelling Multivariate Data with Additive Bayesian Networks
abodOutlier	Angle-Based Outlier Detection

Uporaba paketov

1

```
install.packages("foo")
```

Instalira paket na računalnik

1 x za vsak računalnik

2

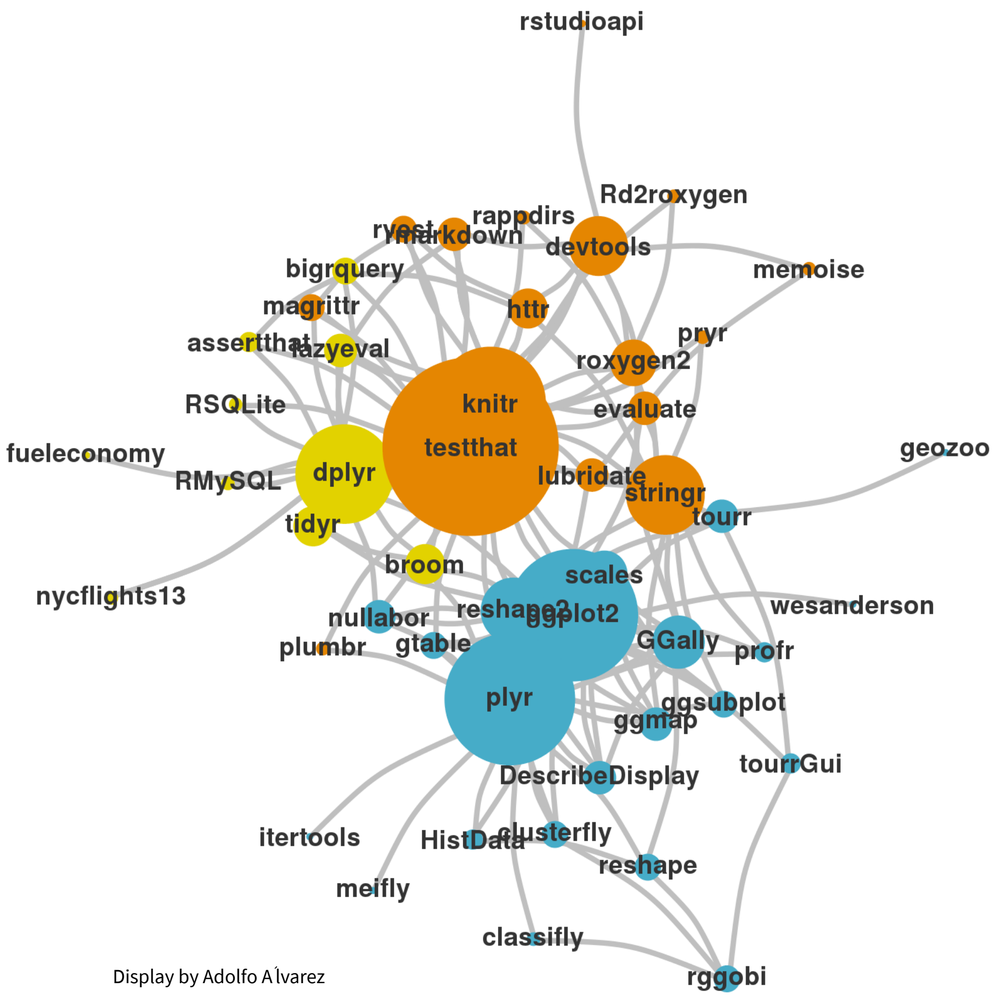
```
library("foo")
```

Naloži paket

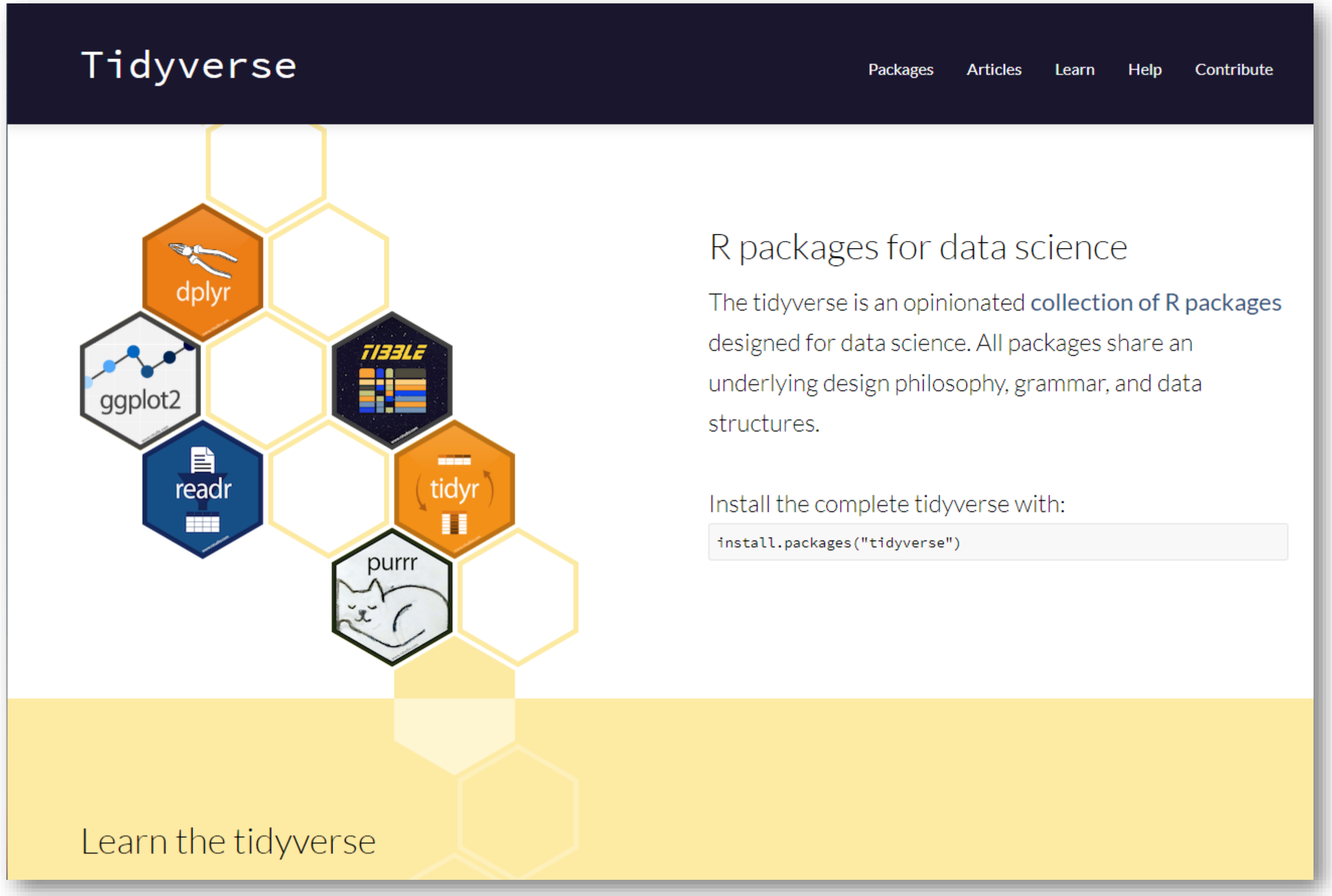
1 x pri vsaki uporabi R-a

“Tidyverse"

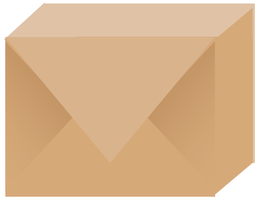
Zbirka sodobnih R paketov, ki si delijo filozofijo, način uporabe in so narejeni, da **delajo skupaj**.



tidyverse.org



tidyverse



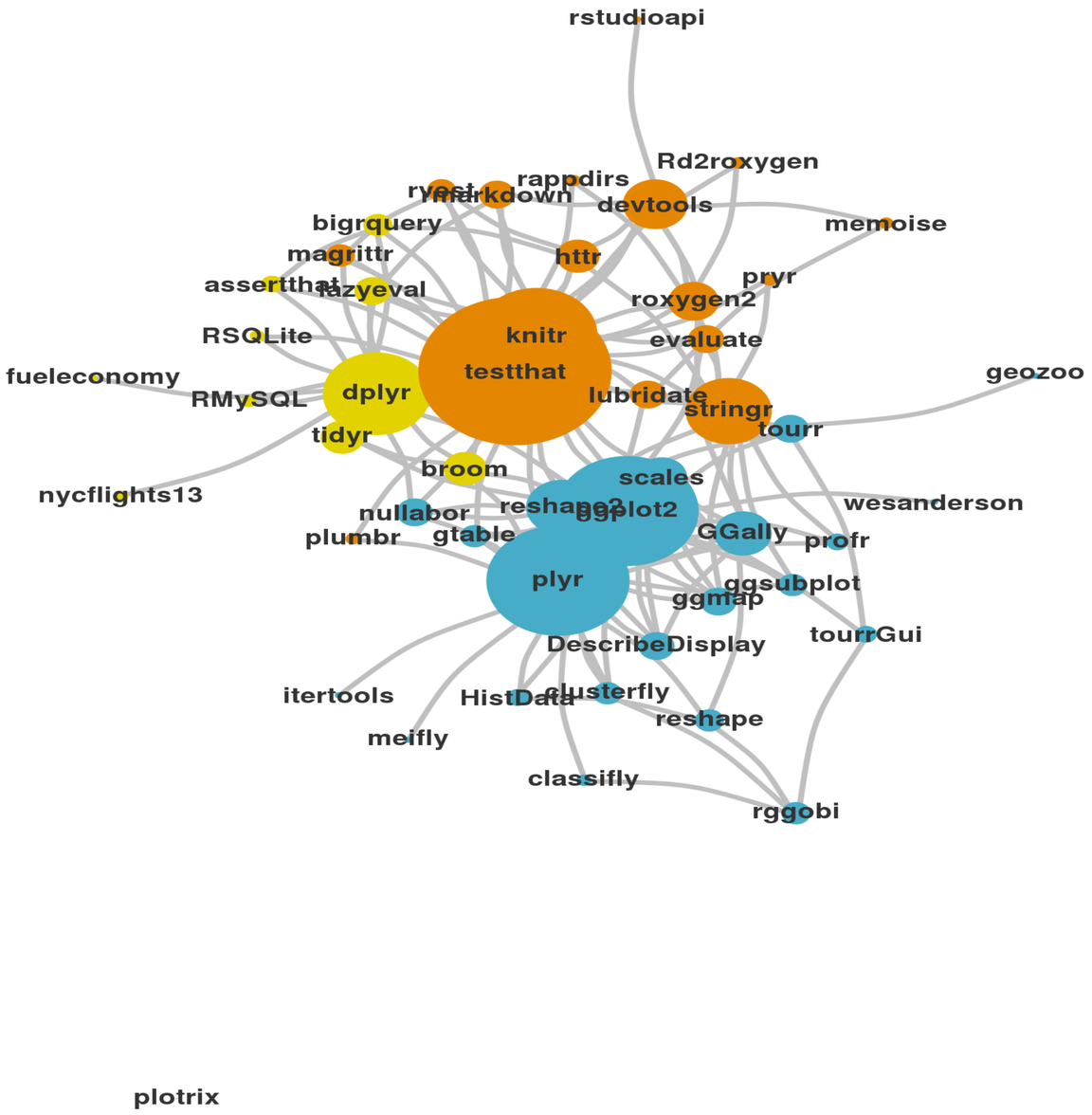
R paket, ki je “bližnjica” za instalacijo in nalaganje paketov iz Tidyverse zbirke.

```
library("tidyverse")
```

```
install.packages("tidyverse")
```

naredi tole:

```
install.packages("ggplot2")
install.packages("dplyr")
install.packages("tidyr")
install.packages("readr")
install.packages("purrr")
install.packages("tibble")
install.packages("hms")
install.packages("stringr")
install.packages("lubridate")
install.packages("forcats")
install.packages("DBI")
install.packages("haven")
install.packages("httr")
install.packages("jsonlite")
install.packages("readxl")
install.packages("rvest")
install.packages("xml2")
install.packages("modelr")
install.packages("broom")
```



```
install.packages("tidyverse")
```

naredi tole:

```
install.packages("ggplot2")
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install.packages("tidyr")
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install.packages("modelr")
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```

```
library("tidyverse")
```

naredi tole:

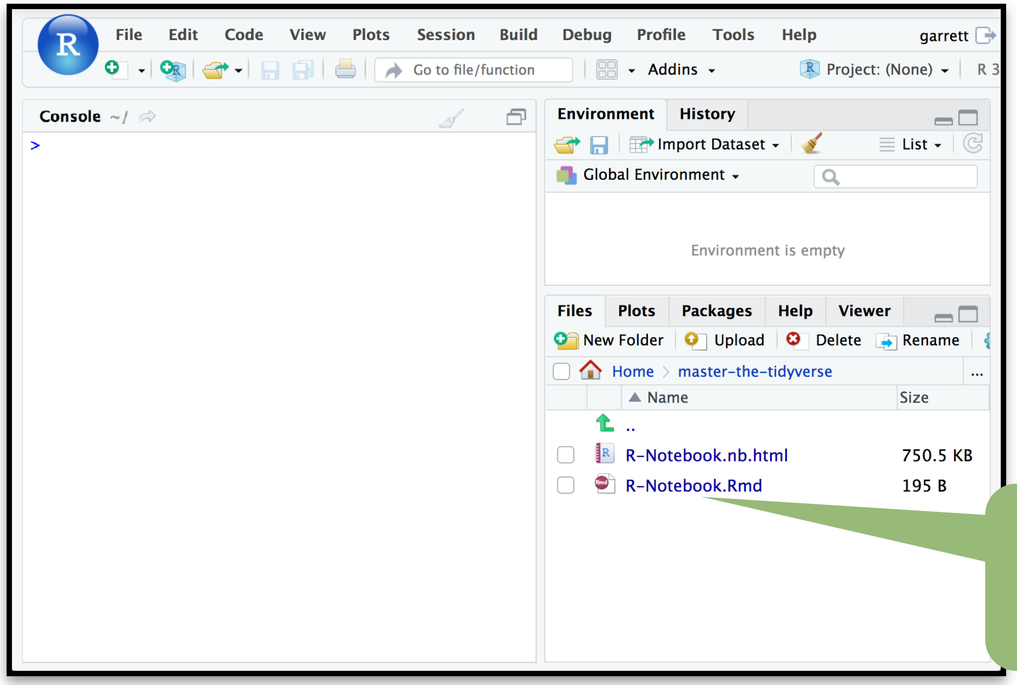
```
library("ggplot2")
library("dplyr")
library("tidyr")
library("readr")
library("purrr")
library("tibble")
```

‘R Notebooks' in ‘Markdown’ (Akcija!)

Orodja

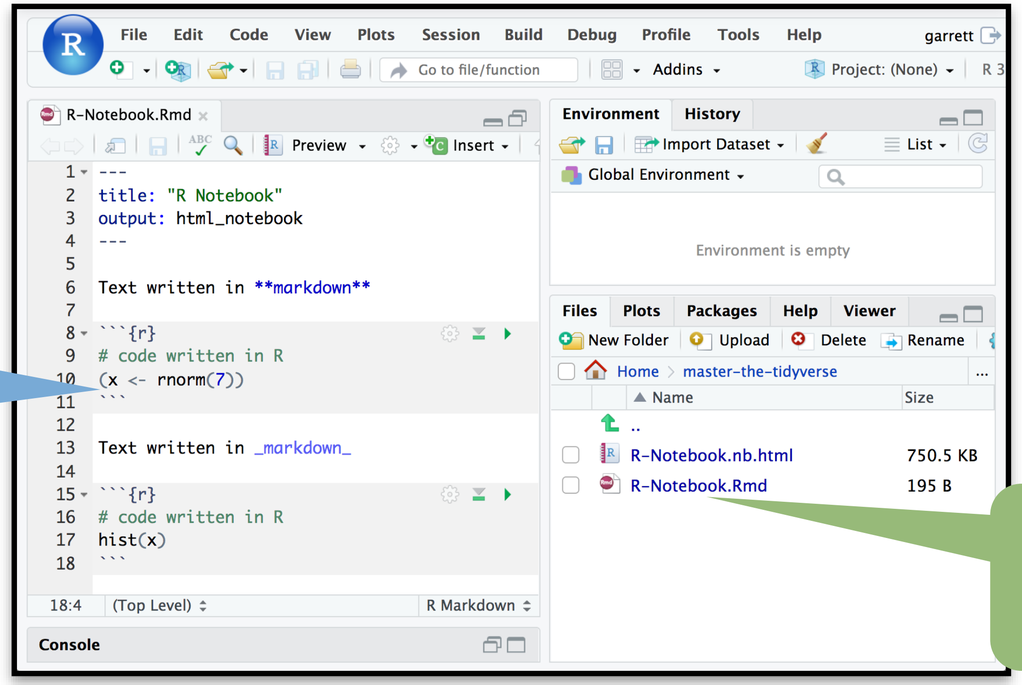
- **<u>Potrebujete</u>**
- R Studio + R
- Paket:
  - tidyverse
- Če nimate instaliranih paketov:
- install.packages(tidyverse”)

Orodja – R Studio in R Notebook



Klikni na.Rmd (R Markdown) datoteko, da odpreš “R Notebook”

Orodja – R Studio in R Notebook



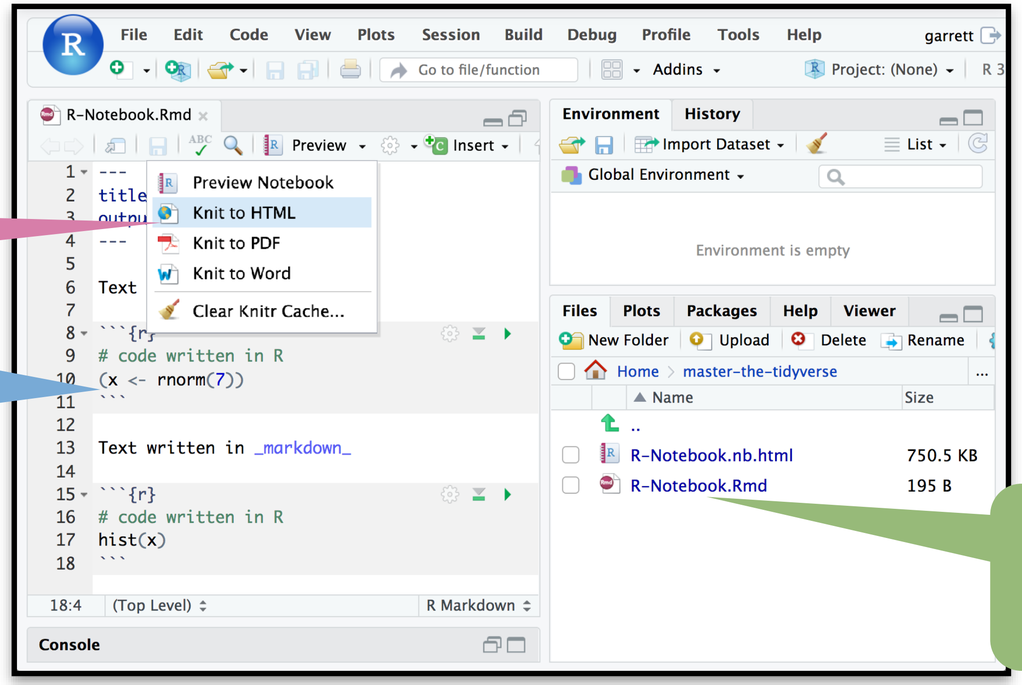
“Notebook” – urejevalnik s kodo in zapiski

Klikni na.Rmd (R Markdown) datoteko, da odpreš “R Notebook”

Orodja – R Studio in R Notebook

Izvozi kot poročilo

“Notebook” – urejevalnik s kodo in zapiski



Klikni na.Rmd (R Markdown) datoteko, da odpreš “R Notebook”

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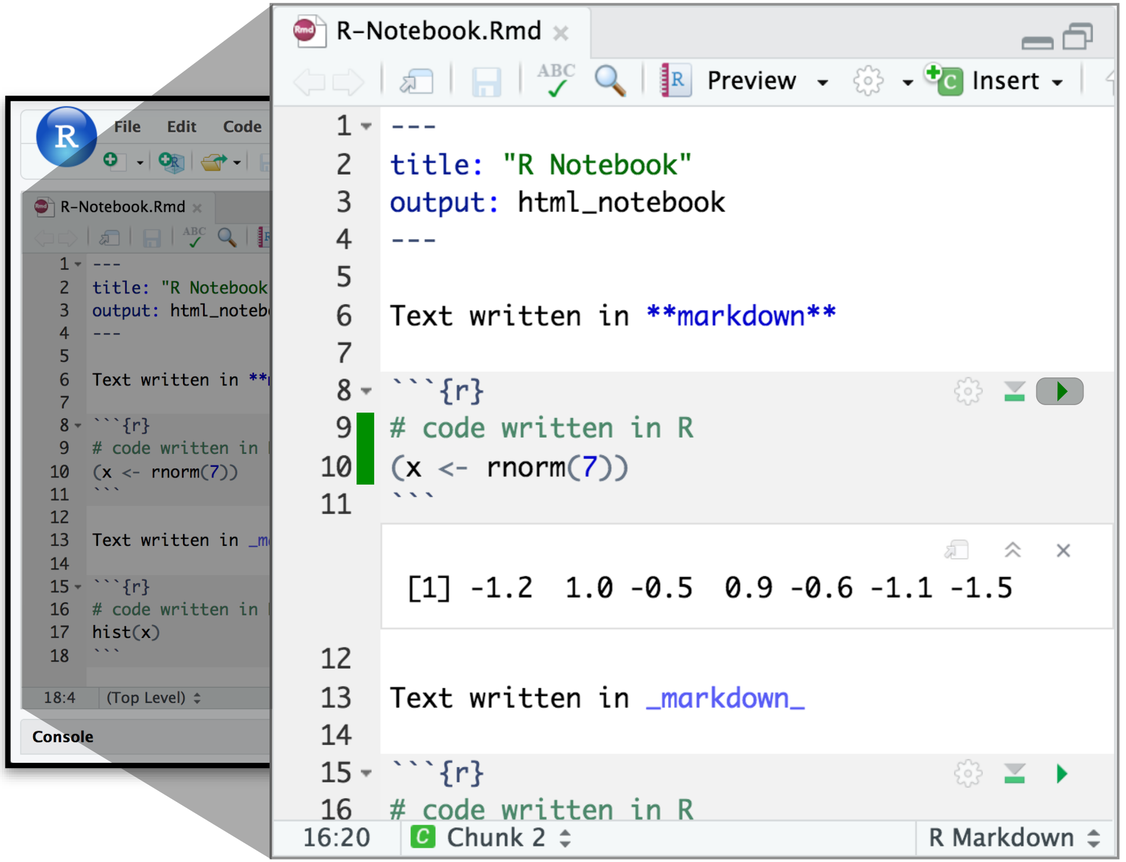
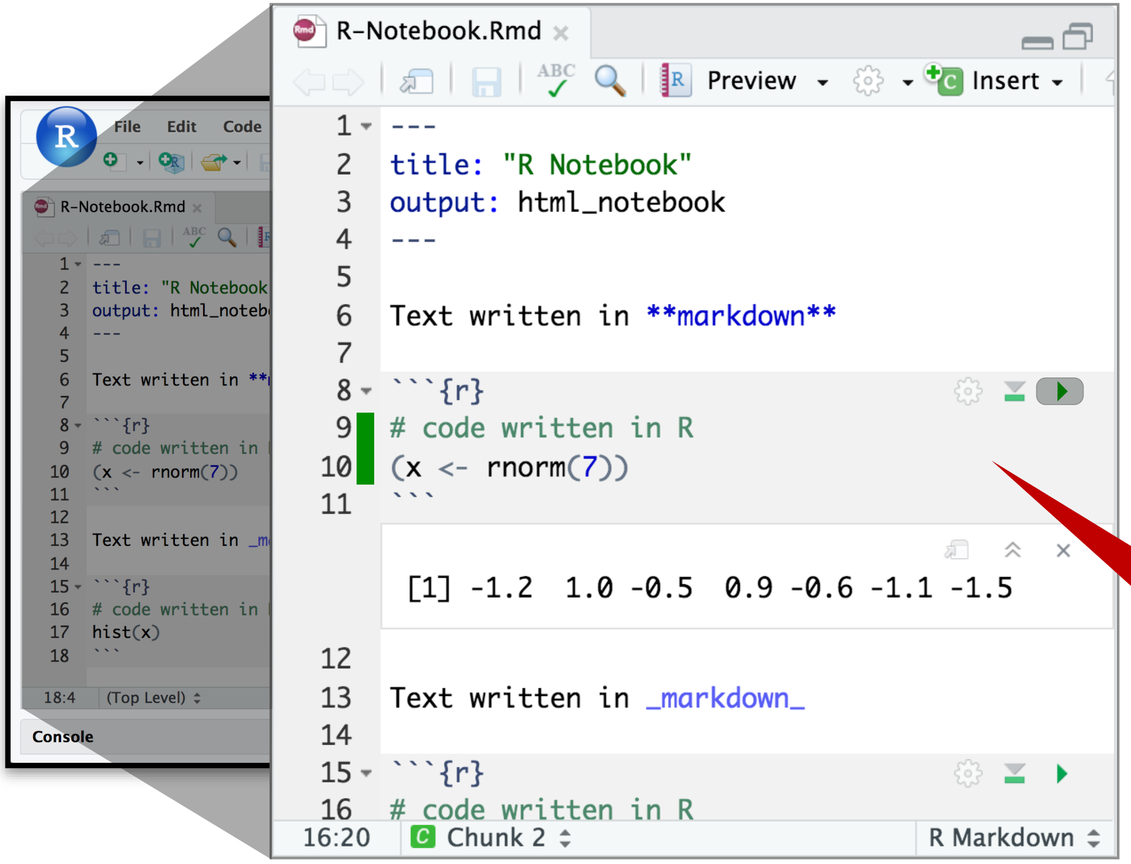
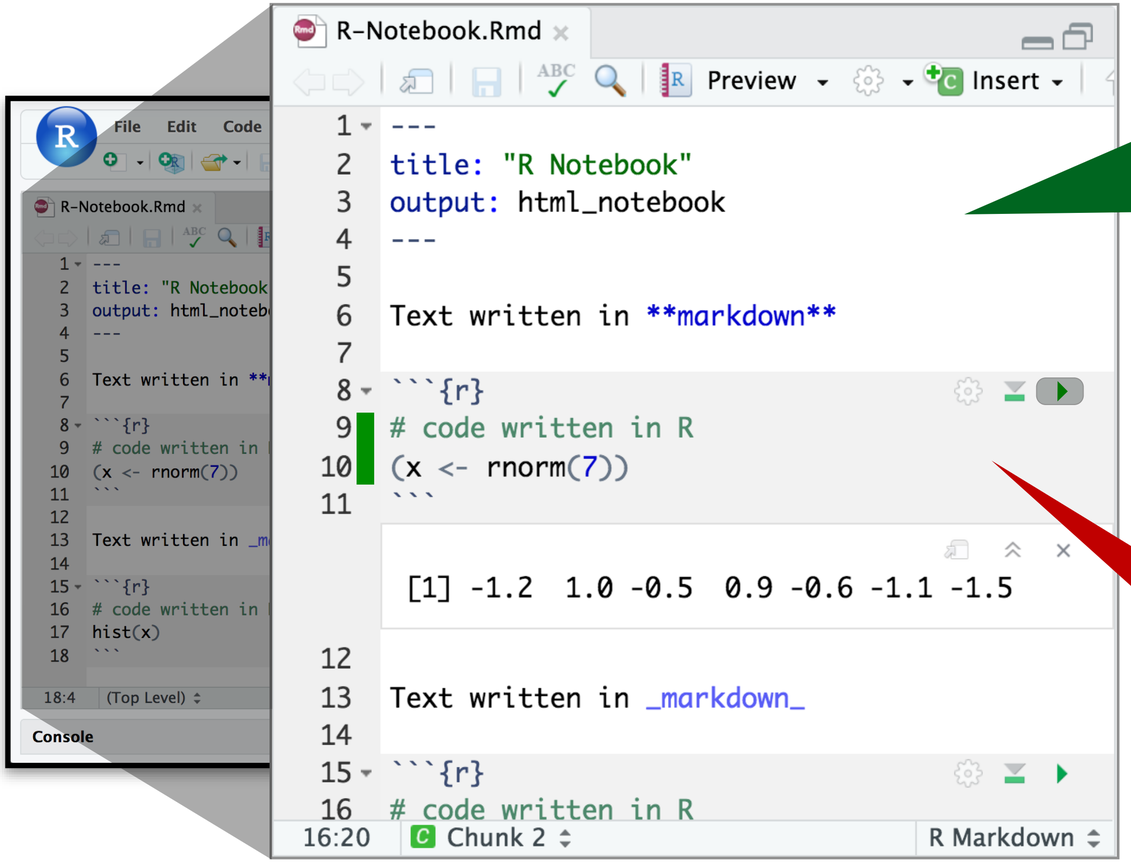


Orodja – R Studio in R Notebook



“Chunk” R kode
(to nekaj naredi)

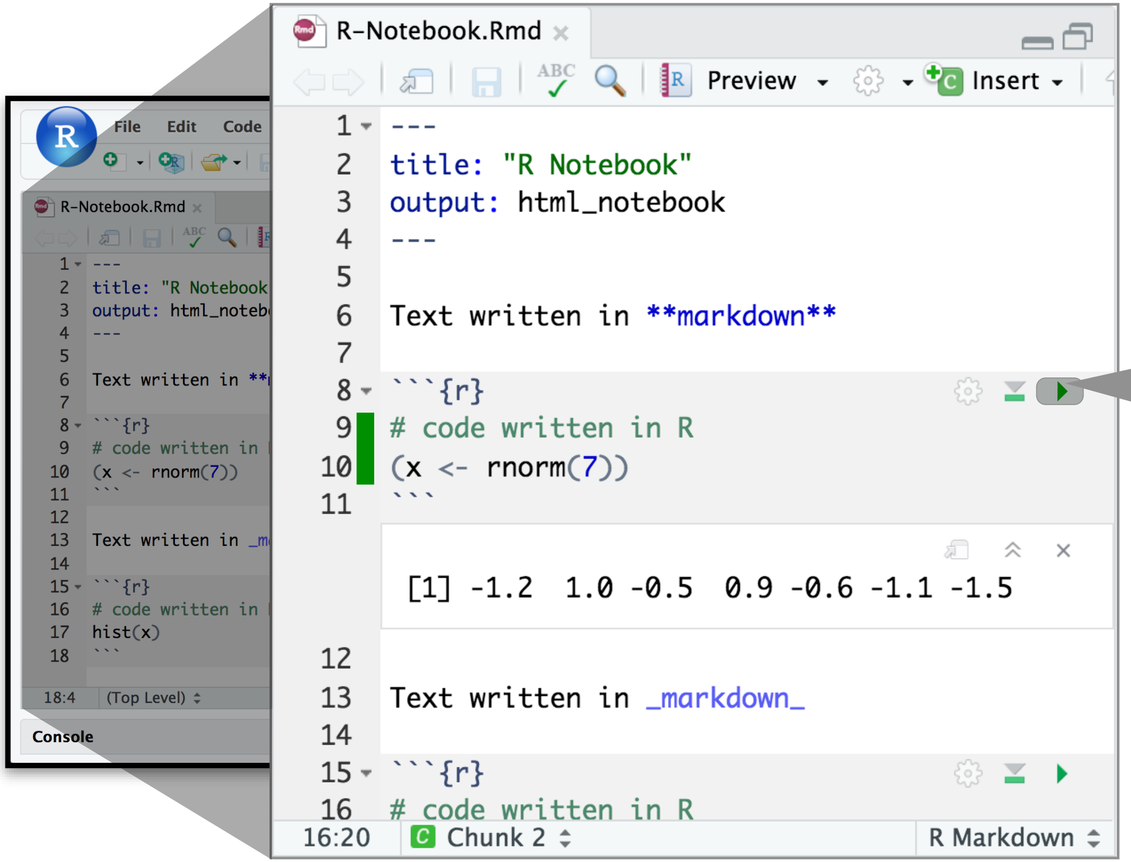
Orodja – R Studio in R Notebook



Zapiski – navadno besedilo, opiše R kodo, naše misli ob kodi

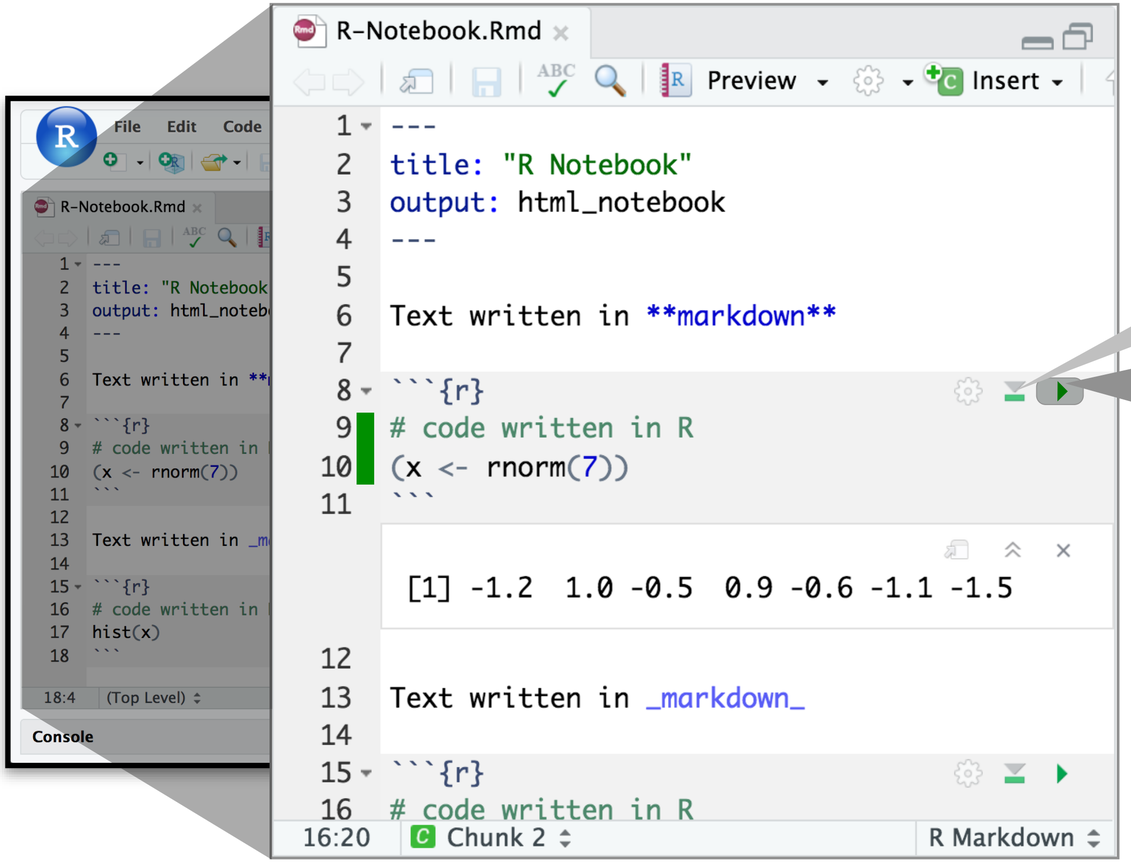
“Chunk” R kode (to nekaj naredi)

Orodja – R Studio in R Notebook



Klikni, da zaženeš kodo v “chunk”-u <ctrl-enter>

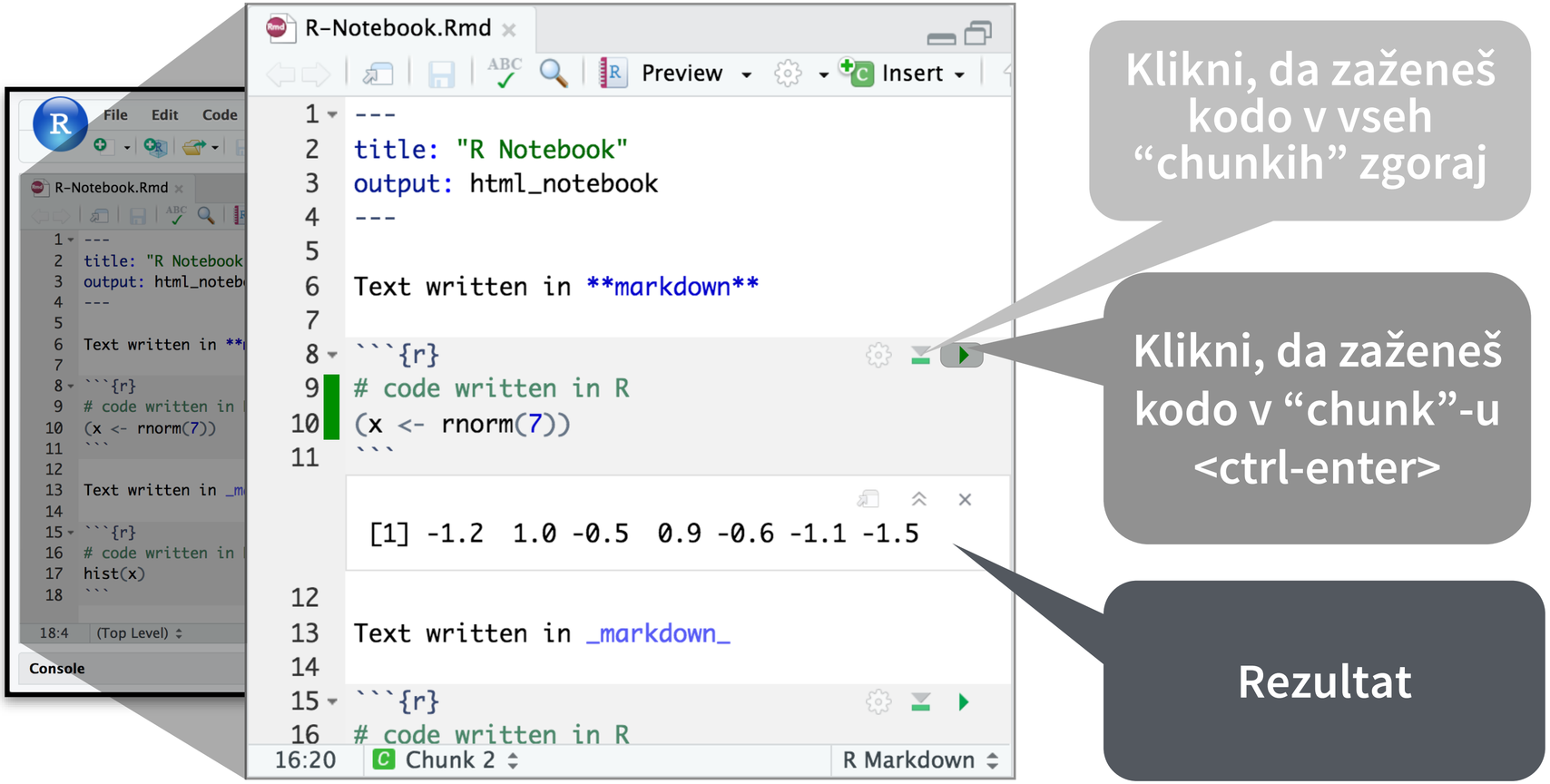
Orodja – R Studio in R Notebook



Klikni, da zaženeš kodo v vseh “chunkih” zgoraj

Klikni, da zaženeš kodo v “chunk”-u <ctrl-enter>

Orodja – R Studio in R Notebook



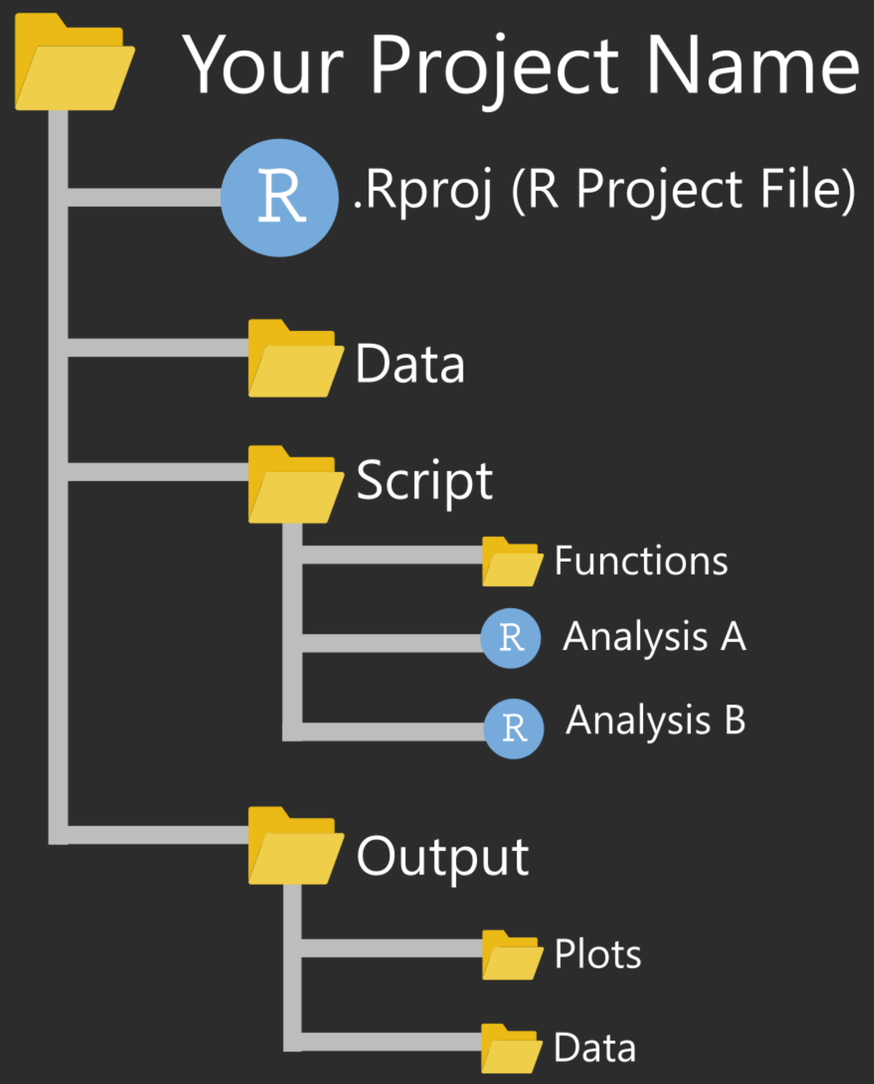
Zdaj pa vi!

Odprite **AVBP_TS_Vaja1.Rproj** (Rstudio project)

V RStudiu odprite **00-Uvod.Rmd**

Rstudio projekt
(.Rproj)

A basic R project set up



Epilog

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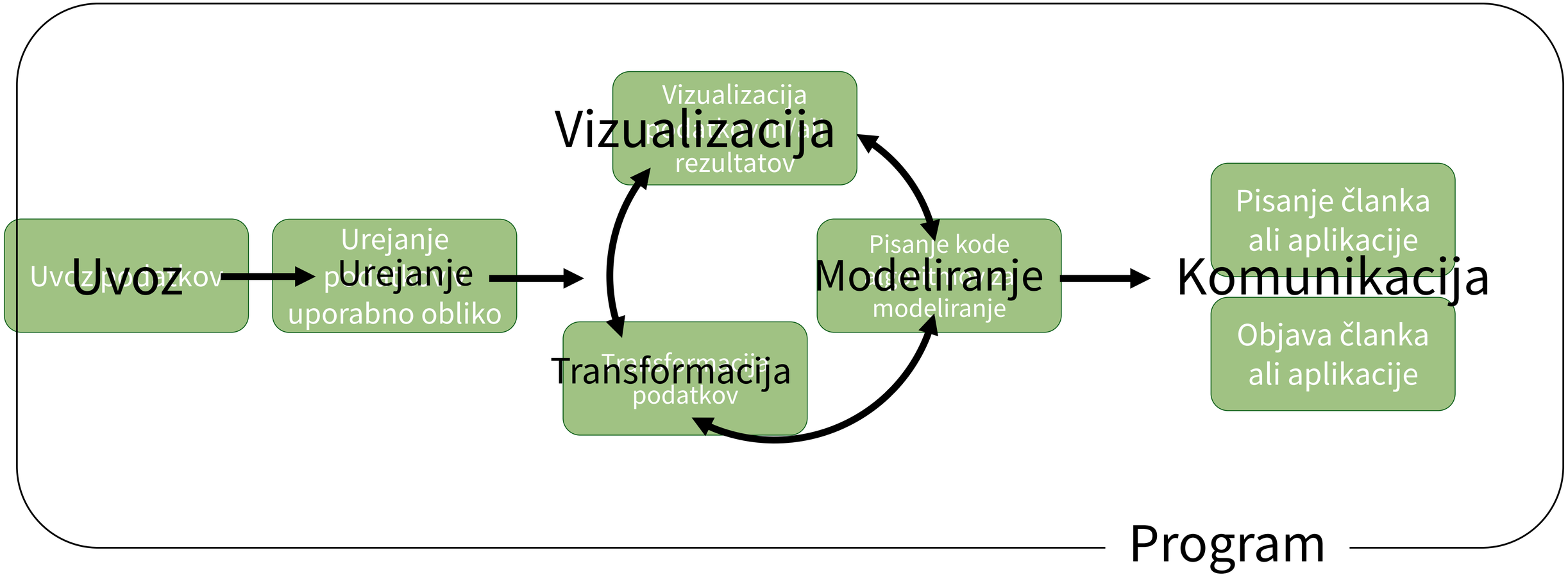
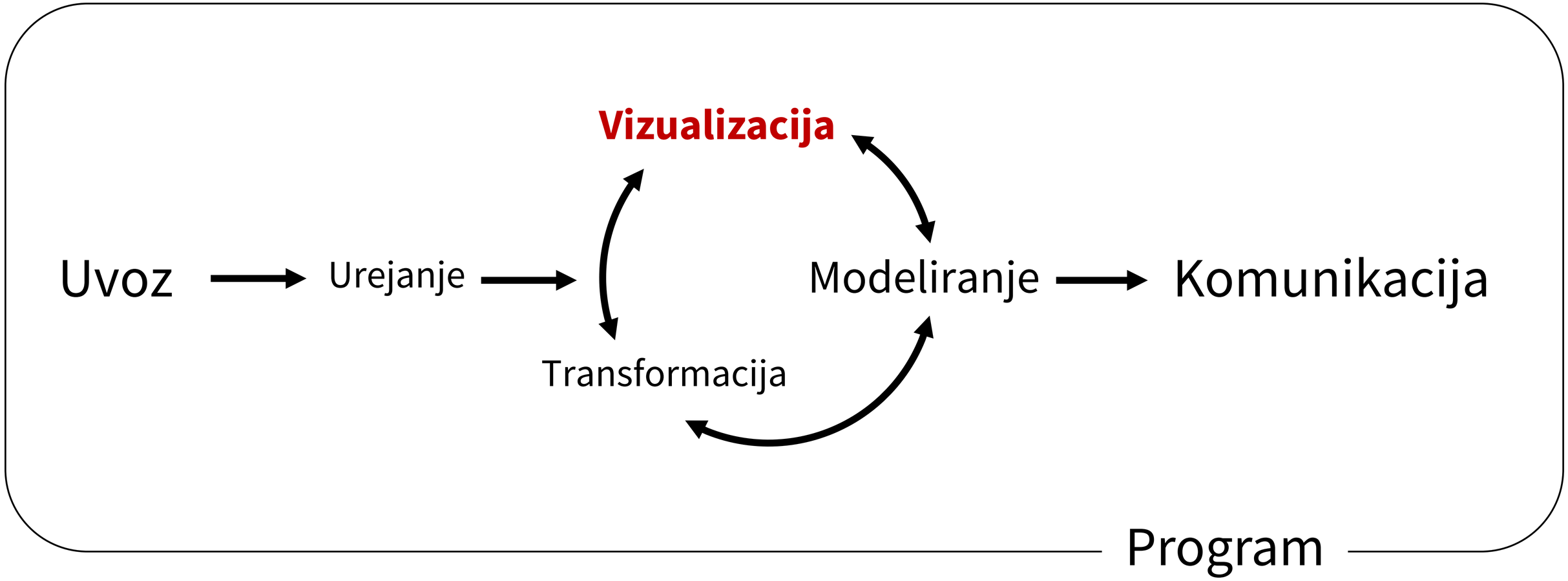
(Uporabna) **podatkovna znanost**



(Uporabna) **podatkovna znanost**



Zdaj pa vi!

Odprite **01-VizualizacijaPodatkov.Rmd**